

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			0.05 mol (2) award (1) for $\frac{5.23}{106}$ or 0.049 if incorrect answer		2		2	2	2
	(b)	(i)	I	to find out the approximate volume (of acid) needed / to get a rough result so as to save time when carrying out accurate titration	1			1		1
			II	sodium carbonate because less volume of it needed both needed			1	1		1
		(ii)		<u>repeat</u> (the titration) <u>without the indicator</u> (1) 25 cm ³ of Na ₂ CO ₃ (1) 27.65 cm ³ of H ₂ SO ₄ (1) [award (1) for same volume of sodium carbonate <u>and</u> mean volume of sulfuric acid] evaporate the water / leave to evaporate / allow to crystallise (1)	4			4		4
	(c)	(i)		CuSO₄ 1 mark + 2NaOH 1 mark → Cu(OH)₂ 1 mark + Na₂SO₄ correct reactants (1) correct products (1) correct balancing (1) – only if reactants and products correct ignore state symbols		3		3		

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		(ii)		sulfate – add <u>barium chloride (solution)</u> and <u>white precipitate</u> (1) sodium – <u>flame test</u> and <u>yellow/orange flame</u> (1) assume ions are implied if correct tests and observations given	2			2		
	(d)			$\begin{array}{ccccccc} \text{Mg}^{2+} & + & \text{CO}_3^{2-} & \rightarrow & \text{MgCO}_3 \\ \underbrace{\hspace{1.5cm}} & & & & \underbrace{\hspace{1cm}} \\ & & & & 1 \text{ mark} \end{array}$ correct reactants (1) correct products (1) ignore state symbols and attempt at balancing		2		2		
				Question 4 total	7	7	1	15	2	8